

LeafOS Brain Model Selection Benchmark

CPU Selection Followed by RTX 4070 Offload Variation

LeafOS / FlowerOS Measurement Harness

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Abstract

Three locally installed brain-model candidates were measured on CPU with three decode trials per model. The highest-throughput CPU candidate was then measured with CPU-only, partial, and full RTX 4070 Vulkan offload. This study separates model selection for the persistent brain lane from hardware acceleration of the selected candidate.

1 Method

All tests used llama.cpp build c4ae9a88f, an Intel i9-13900K, an RTX 4070, a 512-token prompt, a 128-token decode, and three repetitions per setting. CPU selection used `-ngl 0`. GPU variation used the selected model under the same benchmark parameters. The reported value is mean decode throughput in tokens/s.

2 CPU Brain Selection

Table 1: CPU-only decode results.

Candidate	Mean tokens/s	Standard deviation
Qwen3.5 14B A3B MXFP4 MoE	6.269	0.194
Qwopus3.5 9B Q4	4.365	0.039
Gemma4 Opus Q4	4.051	0.029

The selected CPU brain is Qwen3.5 14B A3B MXFP4 MoE. It exceeds the next CPU candidate by

$$\frac{6.269}{4.365} = 1.44 \times .$$

The raw trial results and CPU chart are stored in `reports/brain-selection-20260717/results.json` and `cpu-decode.svg`.

3 GPU Offload Variation of Selected Brain

Full offload produces

$$\frac{107.550}{6.259} = 17.18 \times$$

Table 2: Qwen3.5 14B A3B MXFP4 MoE decode results by GPU placement.

Placement	Mean tokens/s	Standard deviation
CPU only (0 layers)	6.259	0.093
Partial offload (16 layers)	9.917	0.677
Full offload (999 layers)	107.550	0.163

the CPU-only decode throughput, while partial offload provides only

$$\frac{9.917}{6.259} = 1.58 \times .$$

The full-offload result is therefore the preferred interactive/worker configuration when its VRAM requirements are satisfied. The CPU-only rate remains a viable persistent low-rate brain baseline, but it is not the preferred production throughput configuration.

4 Operational Interpretation

LeafOS retains authority, checkpoints, validation, and reporting regardless of model selection. FlowerOS owns the RTX 4070 placement, telemetry, isolation, and reporting surface. GLM-5.2 is now a deprecated historical proposal. This study establishes the measured local comparison baseline for the medium-MoE candidate policy, but its 14B A3B result does not establish that a 47–156B candidate fits the host.

5 Artifacts

- Raw llama.cpp JSON: `reports/brain-selection-20260717/raw/`.
- Normalized data: `reports/brain-selection-20260717/results.json`.
- Visualizations: `cpu-decode.svg` and `gpu-variation.svg` in the same report directory.